Radiation & half-life

Compare emissions

1 Alpha, beta, gamma

Alpha: helium nucleus, highly ionising, stopped by paper/skin; cm in air.

Beta: fast nuclear electron; medium ionisation, stopped by mm of aluminium; metres in air.

Gamma: EM radiation; less ionising, very penetrating; thick lead/concrete reduces intensity.

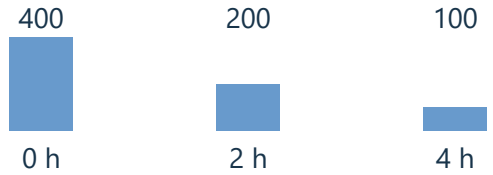
2 Random decay

Cannot predict when one nucleus will decay. A large sample follows a predictable pattern.

Activity = decays/s (Bq). Count rate is detector events per unit time; not all emissions are detected.

Use decay maths

3 Half-life



Count rate halves every 2 h

Same time interval halves the remaining undecayed nuclei or activity.

640 → 320 → 160 → 80 Bq: three half-lives.
With 2 h half-life, elapsed time = **6 h.**

4 Remaining vs decayed

After n half-lives, remaining fraction = **$(½)^n$.**

Three half-lives: remaining 1/8, decline 7/8.
Remaining: initial = 1:8; decline: initial = 7:8.

Measure carefully

5 Background correction

Measure background without source; subtract from total using matching time units.

84 total − 4 background = 80 counts/s source.
One half-life later: 40 + 4 = **44 total.**

6 Graph & measurement

Read time between a source count and half that count; check other starting values.

Keep geometry/shielding fixed. Longer counting and repeats reduce random fluctuations; they do not control decay.